

Three ways to set wireless interface to Monitor mode and Managed mode

You can use the following command to set wireless interface to Monitor mode and Managed mode on any Linux distro. The only requirement is availability wireless adapter that supports monitor mode. [This one](#) is recommended.

1. How to enable monitor mode using iw

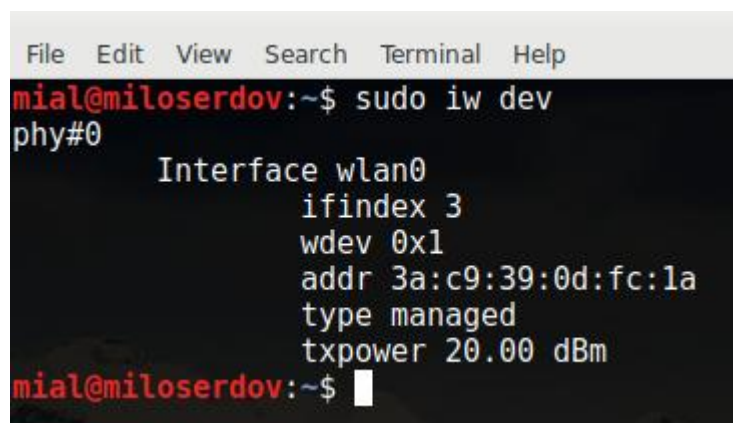
You should check whether the operating system is able to recognize your Wi-Fi card. In addition, you need to know the name of the wireless interface.

Get to know the wireless interface name:

```
1 | sudo iw dev
```

Output:

```
1 | phy#0
2 |     Interface wlan0
3 |         ifindex 3
4 |         wdev 0x1
5 |         addr 3a:c9:39:0d:fc:1a
6 |         type managed
7 |         txpower 20.00 dBm
```

A screenshot of a terminal window with a menu bar (File, Edit, View, Search, Terminal, Help) and a title bar. The terminal shows the command 'sudo iw dev' being executed. The output is: 'phy#0', 'Interface wlan0', 'ifindex 3', 'wdev 0x1', 'addr 3a:c9:39:0d:fc:1a', 'type managed', and 'txpower 20.00 dBm'. The prompt 'mial@miloserdov:~\$' is visible at the bottom.

As you can see, the name of my wireless interface is **wlan0**. In addition, you can see that it is in **managed** mode.

To set wireless interface to Monitor mode with iw you can use the following

command sequence:

```
1 | sudo ip link set IFACE down
2 | sudo iw IFACE set monitor control
3 | sudo ip link set IFACE up
```

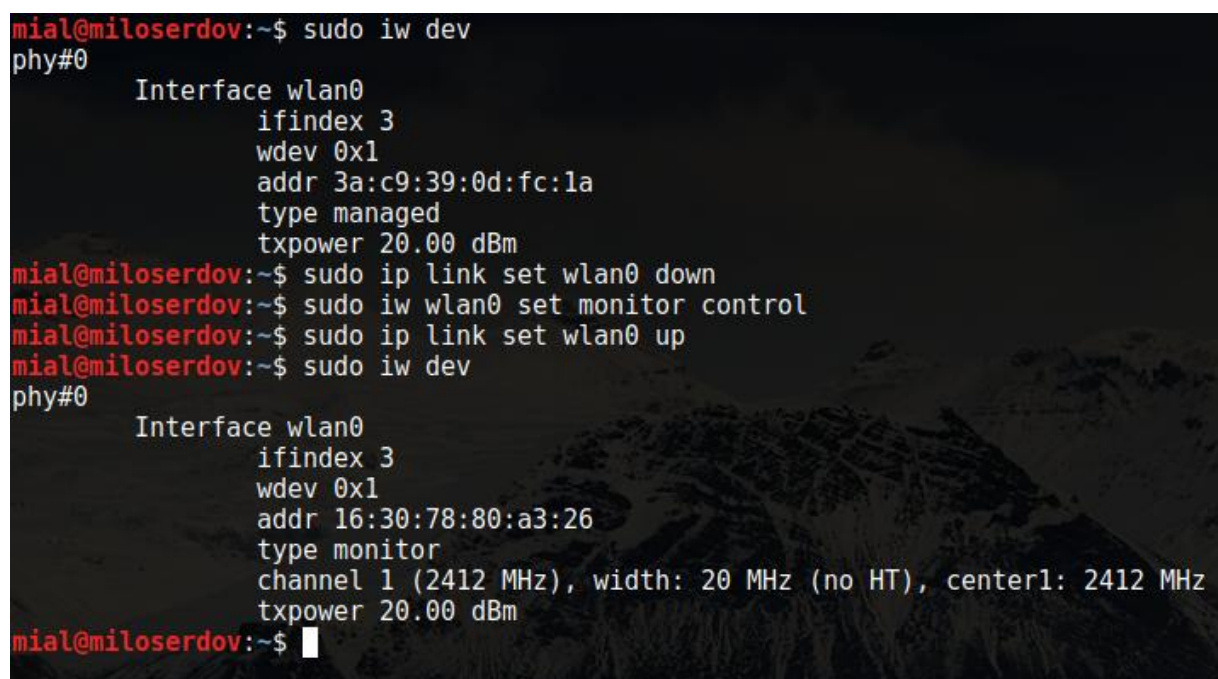
Where **IFACE** replace with actual name of your wireless interface. In may example:

```
1 | sudo ip link set wlan0 down
2 | sudo iw wlan0 set monitor control
3 | sudo ip link set wlan0 up
```

Then check the status of you wireless interface one more time:

```
1 | sudo iw dev
2 | phy#0
3 |     Interface wlan0
4 |         ifindex 3
5 |         wdev 0x1
6 |         addr 16:30:78:80:a3:26
7 |         type monitor
8 |         channel 1 (2412 MHz), width: 20 MHz (no HT), center1:
9 |         txpower 20.00 dBm
```

As you can see, now type **monitor**. Note: the name of interface is not changed by this method.



```
mial@miloserdov:~$ sudo iw dev
phy#0
    Interface wlan0
        ifindex 3
        wdev 0x1
        addr 3a:c9:39:0d:fc:1a
        type managed
        txpower 20.00 dBm
mial@miloserdov:~$ sudo ip link set wlan0 down
mial@miloserdov:~$ sudo iw wlan0 set monitor control
mial@miloserdov:~$ sudo ip link set wlan0 up
mial@miloserdov:~$ sudo iw dev
phy#0
    Interface wlan0
        ifindex 3
        wdev 0x1
        addr 16:30:78:80:a3:26
        type monitor
        channel 1 (2412 MHz), width: 20 MHz (no HT), center1: 2412 MHz
        txpower 20.00 dBm
mial@miloserdov:~$
```

To return wireless interface in Managed mode with **iw** you can use the following command sequence:

```
1 | sudo ip link set IFACE down
2 | sudo iw IFACE set type managed
3 | sudo ip link set IFACE up
```

Where **IFACE** replace with actual name of your wireless interface. In may example:

```
1 | sudo ip link set wlan0 down
2 | sudo iw wlan0 set type managed
3 | sudo ip link set wlan0 up
```

2. How to enable monitor mode using Airmmon-ng

Again, we should get information about our wireless interface:

```
1 | sudo airmon-ng
```

Output:

```
1 | PHY Interface  Driver      Chipset
2 |
3 | phy0      wlan0      rt2800usb  Ralink Technology, Corp. RT35
```

The name of interface is **wlan0**.

Checking for interfering processes

Before putting a card into monitor mode, it will automatically check for interfering processes. It can also be done manually by running the following command:

```
1 | sudo airmon-ng check
```

This command stops network managers then kill interfering processes left:

```
1 | sudo airmon-ng check kill
```

At last, we **start monitor mode**:

```
1 | sudo airmon-ng start wlan0
2 |
3 |
4 | PHY Interface  Driver      Chipset
5 |
6 | phy0      wlan0      rt2800usb  Ralink Technology, Corp. RT35
7 |
8 |              (mac80211 monitor mode vif enabled for [phy0]wlan0 on
9 |              (mac80211 station mode vif disabled for [phy0]wlan0)
```

As you can see, it created a monitor mode interface called **wlan0mon**.

```
1  sudo iwconfig
2  wlan0mon IEEE 802.11 Mode:Monitor Frequency:2.457 GHz Tx-
3      Retry short limit:7 RTS thr:off Fragment thr:of
4      Power Management:off
5
6  lo      no wireless extensions.
7
8  eth0    no wireless extensions.
```

```
mial@miloserdov:~$ sudo airmon-ng

PHY      Interface      Driver      Chipset
phy0     wlan0             rt2800usb   Ralink Technology, Corp. RT3572

mial@miloserdov:~$ sudo airmon-ng check

Found 3 processes that could cause trouble.
If airodump-ng, aireplay-ng or airtun-ng stops working after
a short period of time, you may want to run 'airmon-ng check kill'

  PID Name
  544 NetworkManager
  635 dhclient
 1355 wpa_supplicant

mial@miloserdov:~$ sudo airmon-ng check kill

Killing these processes:

  PID Name
  635 dhclient
 1355 wpa_supplicant

mial@miloserdov:~$ sudo airmon-ng start wlan0

PHY      Interface      Driver      Chipset
phy0     wlan0             rt2800usb   Ralink Technology, Corp. RT3572

              (mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)
              (mac80211 station mode vif disabled for [phy0]wlan0)

mial@miloserdov:~$ sudo iwconfig
wlan0mon IEEE 802.11 Mode:Monitor Frequency:2.457 GHz Tx-Power=20 dBm
      Retry short limit:7 RTS thr:off Fragment thr:off
      Power Management:off

lo      no wireless extensions.

eth0    no wireless extensions.

mial@miloserdov:~$
```

Disable monitor mode

```

1 | sudo airmon-ng stop wlan0mon
2
3 | PHY Interface      Driver      Chipset
4
5 | phy0      wlan0mon      rt2800usb      Ralink Technology, Corp. RT35
6
7 |                (mac80211 station mode vif enabled on [phy0]wlan0)
8
9 |                (mac80211 monitor mode vif disabled for [phy0]wlan0mo

```

Don't forget to restart the Network Manager. It is usually done with the following command:

```

1 | sudo systemctl start NetworkManager

```

3. How to enable monitor mode using iwconfig

As usual, start from checking interface name:

```

1 | sudo iwconfig
2 | lo          no wireless extensions.
3
4 | eth0        no wireless extensions.
5
6 | wlan0       IEEE 802.11  ESSID:off/any
7 |              Mode:Managed  Access Point: Not-Associated  Tx-Po
8 |              Retry short limit:7  RTS thr:off  Fragment thr:o
9 |              Encryption key:off
10 |             Power Management:off

```

The network interface with wireless extension is called **wlan0**.

Enable monitor mode:

```

1 | sudo ifconfig IFACE down
2 | sudo iwconfig IFACE mode monitor
3 | sudo ifconfig IFACE up

```

Actual example:

```

1 | sudo ifconfig wlan0 down
2 | sudo iwconfig wlan0 mode monitor
3 | sudo ifconfig wlan0 up

```

Disable monitor mode:

```

1 | sudo ifconfig wlan0 down
2 | sudo iwconfig wlan0 mode managed
3 | sudo ifconfig wlan0 up

```

NetworkManager prevents monitor mode

If NetworkManager restarts automatically after each kill, and it pretends monitor mode, you can stop it manually:

In Kali Linux, BlackArch, Ubuntu, Linux Mint:

```
1 | sudo systemctl stop NetworkManager
```

Note: when you stop NetworkManager, your Internet access disappears!

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P says:

June 27, 2019 at 12:17 am

At sudo ifconfig IFACE down: IFACE: ERROR while getting interface flags: No such device

[Reply](#)

monyettenyom says:

December 3, 2019 at 10:29 am

I'm trying to enable the mode monitor in the wireless adapter. I did these:

```
“ sudo ip link set wlan0 down sudo iw wlan0 set monitor none sudo ip  
link set wlan0 up ”
```

And its alternative:

```
“ sudo ifconfig wlan0 down sudo iwconfig wlan0 mode monitor sudo  
ifconfig wlan0 up ”
```

It worked for 2 seconds, but the wireless adapter restarted and change its mode to managed again before I was able to perform the next steps for packet forwarding, arpspoofing and capture packets with Wireshark.

After those steps above, I've tried sudo network-manager stop to stop nw to restart, but then I can't capture TCP/UCP/HTTP with Wireshark as I wished. Wireshark captured only WLAN packets.

I'm using Kali Rolling 2019.4 in Lenovo E320.

What did I do wrong? Help!

[Reply](#)

Anonymous says:

December 19, 2019 at 8:57 pm

guys all the three methods works on my system man ! great work its easy to go on monitor mode

[Reply](#)

Tazul Islam says:

March 7, 2022 at 11:40 am

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